Student Number: | | | | | | | | | |

NATIONAL UNIVERSITY OF SINGAPORE

MA1522 - Linear Algebra for Computing

(Semester 2 : AY2024/2025)

Time allowed : 2 hours

INSTRUCTIONS TO CANDIDATES

1. **Write down your matriculation/student number clearly in the space provided at the top of this page.** Do not write your name.
2. This (and only this) booklet will be collected at the end of the examination.
3. This examination paper contains **6** questions and comprises **35** printed pages.
4. Answer **ALL** questions. The maximum possible marks are **100**.
5. This is a CLOSED BOOK (with helpsheet) examination.
6. Students are allowed to use one double-sided A4 size helpsheet.
7. Candidates may use non-programmable calculators. However, they should lay out systematically the various steps in the calculations.

For official use only. Do not write below this line.

Question	1	2	3	4	5	6	Total
Marks							

Indicate whether each of the following statements is true. No justification is required.

(a) The linear system $\mathbf{Ax} = \mathbf{0}$ is always consistent.

Please answer as follows: ☐ TRUE, ☐ FALSE.

(b) If two matrices are row equivalent, then their column spaces are the same.

Please answer as follows: ☐ TRUE, ☐ FALSE.

(c) If $\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\}$ is a basis for a subspace V of $\mathbb{R}^n$, then so is

$$\{\mathbf{u}_1 + 2\mathbf{u}_2, \mathbf{u}_2 - \mathbf{u}_3, 2\mathbf{u}_1 + \mathbf{u}_2 + 3\mathbf{u}_3\}.$$

Please answer as follows: ☐ TRUE, ☐ FALSE.

(d) Let $\mathbf{A}$ be a 3×4 matrix with LU-factorization

$$\mathbf{A} = \mathbf{L} \begin{pmatrix} 1 & 3 & 1 & -2 \\ 0 & 0 & 3 & 1 \\ 0 & 0 & 0 & 5 \end{pmatrix},$$

and $\mathbf{a}_i$ be the i-th column of $\mathbf{A}$, for $i = 1, 2, 3, 4$. Then $\{\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3\}$ is a basis for the column space of $\mathbf{A}$.

Please answer as follows: ☐ TRUE, ☐ FALSE.

(e) Any symmetric square matrix is orthogonally diagonalizable.

Please answer as follows: ☐ TRUE, ☐ FALSE.

(f) If $\mathbf{Q}_1$ and $\mathbf{Q}_2$ are orthogonal matrices, then $\mathbf{Q}_1\mathbf{Q}_2$ is also an orthogonal matrix.

Please answer as follows: ☐ TRUE, ☐ FALSE.

(g) If an $m \times n$ matrix $\mathbf{B}$ has linearly dependent columns and $\mathbf{b}$ is a vector in $\mathbb{R}^m$, then $\mathbf{b}$ does not have a unique projection onto the column space of $\mathbf{B}$.

Please answer as follows: ☐ TRUE, ☐ FALSE.

(h) Let $S = \{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\}$ be an orthonormal basis for a subspace V of $\mathbb{R}^n$. Then for any $\mathbf{x}, \mathbf{y}$ in V, $[\mathbf{x}]_S \cdot [\mathbf{y}]_S = \mathbf{x} \cdot \mathbf{y}$.

Please answer as follows: ☐ TRUE, ☐ FALSE.

(i) If a matrix $\mathbf{A}_{n\times n}$ satisfies $\mathbf{A}^{2025} = \mathbf{0}_n$ then the eigenvalues of $\mathbf{A}$ can only be 0.

Please answer as follows: ☐ TRUE, ☐ FALSE.

(j) Let $T : \mathbb{R}^n \to \mathbb{R}^n$ be a linear transformation. If $T(\mathbf{x}_1) = T(\mathbf{x}_2)$, then $\mathbf{x}_1 = \mathbf{x}_2$.

Please answer as follows: ☐ TRUE, ☐ FALSE.

Question 2(i) [5 marks]

Let

$$\mathbf{A} = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \\ 0 & 0 & 1 \end{pmatrix} \quad \text{and} \quad \mathbf{b} = \begin{pmatrix} 6 \\ 15 \\ 24 \\ 1 \end{pmatrix}.$$

Using Gaussian Elimination method to solve the linear system

$$\mathbf{Ax} = \mathbf{b}.$$

You need to show the reductions. Finding answers using any other method will not fetch any marks.

Let $\mathbf{A}$ be as in part (i). Find the reduced row echelon form of $\mathbf{A}$. You do not need to show the reductions.

Let $\mathbf{A}$ be as in part (i). Find a basis of the column space of $\mathbf{A}$ and the rank of $\mathbf{A}$. You do not need to justify the answers.

Let $\mathbf{A}$ be as in part (i) and $U = \{(1,2,3)^T, (4,5,6)^T, (7,8,9)^T, (0,0,1)^T\}$ be the set of its row vectors. Find a subset of U which forms a basis of the row space of $\mathbf{A}$. You need to explain briefly how you obtain the answers.

Question 3(i) [10 marks]

Let

$$\mathbf{A} = \begin{pmatrix} 1 & -2 & -1 \\ 2 & 0 & 1 \\ 2 & -4 & 2 \\ 4 & 0 & 0 \end{pmatrix}.$$

Find an orthonormal basis for the column space of $\mathbf{A}$.

Let $\mathbf{A}$ be as in part (i). Find a QR factorization of $\mathbf{A}$.

Let $\mathbf{A}$ be as in part (i). Find a least squares solution to $\mathbf{Ax} = \mathbf{b}$, where

$$\mathbf{b} = \begin{pmatrix} -1 \\ 1 \\ 1 \\ -2 \end{pmatrix}.$$

Question 4(i) [5 marks]

Let $\mathbf{A} = \begin{pmatrix} a & 1 \\ 1 & a \end{pmatrix}$ where a is a fixed real number. Find all eigenvalues and their associated eigenspaces respectively.

Let $\mathbf{A}$ be as in part (i). Find an invertible matrix $\mathbf{P}$, its inverse $\mathbf{P}^{-1}$, and a diagonal matrix $\mathbf{D}$ such that $\mathbf{A} = \mathbf{PDP}^{-1}$. Explain how you find $\mathbf{P}^{-1}$.

Let $\mathbf{A}$ be as in part (i). Find $\mathbf{A}^m$, where m is a positive integer.

Question 5(i) [5 marks]

Let $T : \mathbb{R}^{2024} \to \mathbb{R}^{2024}$ be defined by

$$T\left(\begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_{2023} \\ x_{2024} \end{pmatrix}\right) = \begin{pmatrix} x_1 + x_2 \\ x_2 + x_3 \\ \vdots \\ x_{2023} + x_{2024} \\ x_{2024} + x_1 \end{pmatrix},$$

that is, the i-th entry in $T(\mathbf{x})$ is $x_i + x_{i+1}$ for $1 \leq i \leq 2023$ and the last entry is $x_{2024} + x_1$. Verify that T is a linear transformation.

Let T be as in (i). Find the standard matrix of T. You do not need to justify your answer.

Let T be as in (i). Find the kernel and the range of T respectively. You do not need to justify your answer.

Let

$$\mathbf{A} = \begin{pmatrix} 1 & 1 \\ 1 & 1 \\ 0 & 0 \end{pmatrix}.$$

Find a singular value decomposition of $\mathbf{A}$.

Please indicate the question number clearly.

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Please indicate the question number clearly.

END OF QUESTION PAPER.